Cofunction identities

1 For each of the following acute angles, state the complementary angle:

a 34°

b 62°

c $\dfrac{\pi}{6}$

d $\dfrac{2\pi}{7}$

2 Write each of the following trigonometric ratios in terms of its cofunction:

a $\cos 37^\circ$

b $\sin 76^\circ$

c $\tan 68^\circ$

d $\cot (43^\circ)$

e $\operatorname{cosec} 35^\circ$

f $\cos \dfrac{\pi}{9}$

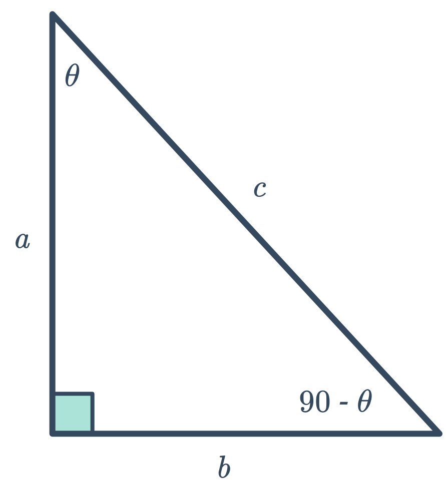
g $\sin \dfrac{\pi}{10}$

h $\cot \left(\dfrac{\pi}{9}\right)$

i $\sec \dfrac{\pi}{9}$

j $\cos \dfrac{1}{5}$

3 Consider the following right triangle:



a Find an expression for the following:

i $\cos \theta$

ii $\sin (90^\circ - \theta)$

iii $\sin \theta$

iv $\cos (90^\circ - \theta)$

b Describe the rule found in part (a) using words.

c If $\sin \alpha = 0.32$, find $\cos (90^\circ - \alpha)$.

d If $\sin \alpha = \cos \beta$, find $\alpha + \beta$.

4 Simplify:

a $\sin (90^\circ - p)$

b $\dfrac{\cos (90^\circ - x)}{\cos x}$

c $\dfrac{\cos 26^\circ}{7 \sin 64^\circ}$

d $\dfrac{\sin 51^\circ}{\cos 39^\circ}$

e $\sin (90^\circ - y) \times \tan y$

5 Given that $\sin x = 0.19$, find the exact value of $\cos \left(\dfrac{\pi}{2} - x\right)$.

6 Given that $\sin x = 0.46$, find the exact value of $\cos (90^\circ - x)$.

7 Prove the following:

a $\dfrac{\sin x \cos (90-x)}{\cos x \sin (90-x)}=\tan^2(x)$

b $\dfrac{\sin x \sin (90-x)}{\cos x \cos (90-x)}=1$

Trigonometric equations

8 Solve for the measure of the acute angle θ in the following equations:

a $\sin\theta=\cos 25^\circ$

b $\cos\theta=\sin 85^\circ$

c $\cos\theta=\sin\dfrac{\pi}{8}$

d $\sin\dfrac{2\pi}{5}=\cos\theta$

9 Solve for the exact value of c, if $\sin\left(\dfrac{\pi}{9}+c\right)=\cos\left(\dfrac{\pi}{4}\right)$.

10 Solve for the value of c, if $\sin(34^\circ+c^\circ)=\cos 51^\circ$.

11 Solve for the value of B if $\cos(66^\circ-B)=\sin 38^\circ$.

12 Solve for the value of A if $\sin\left(\dfrac{A}{68}\right)=\cos 18^\circ$.

13 Solve for the value of x in the following equations:

a $\sin(5x+40^\circ)=\cos(3x+10^\circ)$

b $\sin\left(6x+\dfrac{\pi}{3}\right)=\cos\left(4x+\dfrac{\pi}{9}\right)$